

Boundary-Mode Topographic Cosmology: A Reduced-Response Bridge from Berger–Hopf Geometry to Late-Time Structure Growth

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Abstract

We develop a late-time cosmological bridge between a scalar topographic effective field theory on compact spatial sections $S^3(R_H)$ and the constraint-reduced response architecture of the Berger–Hopf program. A closed-FLRW gauge audit changes the identification of the relevant infrared carrier: the first nonconstant scalar harmonic in our convention, $n = 1$, maps to the exceptional $N = 2$ sector of standard closed-universe perturbation theory and is nondynamical or pure gauge in the regular single-field covariant scalar sector. The first propagating scalar harmonic is therefore our $n = 2$ ($N = 3$ in the alternative convention). This does not remove the sky-dipole mechanism. A general degree-two scalar harmonic on S^3 contains observer-sky monopole, dipole, and quadrupole pieces, and a pure cross-term subspace produces an exact sky dipole proportional to $\sin[2\psi(z)] \hat{\mathbf{n}} \cdot \hat{\mathbf{p}}$. The reduced Schur/Feshbach response can selectively soften this first physical harmonic while stiffening all propagating $n \geq 3$ modes when $1/15 < \ell^2 < 1/8$. Reference Branch R1 and its frozen numerical manifest are left unchanged because the background and high- n growth calculation do not depend on the identity of the curvature-scale carrier. The theoretical harmonic identification is amended prospectively before comparison with the targeted independent data. The resulting framework is falsifiable through curvature, the fixed dipole axis and radial envelope, compact-harmonic hierarchy, growth, and lensing tests.

1 Introduction

Late-time cosmological tensions and directional residuals motivate tests that are capable of separating local structure, survey geometry, and genuinely new large-scale dynamics. The scalar-topographic program introduced a compact hyperspherical parent $S^3(R_H)$ and a long-wavelength scalar response whose largest observational effect occurs at low redshift [1]. A later covariant formulation placed the topographic scalar in the cubic-Galileon sector of Horndeski theory, preserving second-order time evolution while allowing nontrivial late-time scalar–metric braiding.

The present work asks a narrower question: can the long-wavelength topographic response be represented as the large-scale branch of the same constraint-reduced response logic used in the Berger–Hopf program, without assuming that the full microscopic BHSM construction has already been compactified into cosmology? We answer this at the level of a controlled bridge model.

The central mathematical chain is

$$S[\Phi] \longrightarrow \hat{\mathcal{H}}_{\text{phys}} \longrightarrow \mathcal{K}_{\text{topo}} \longrightarrow \lambda_{\text{topo}} \longrightarrow T_n \longrightarrow \delta g_{\mu\nu} \longrightarrow \{D_L, H, f\sigma_8, \Phi + \Psi\}.$$

The distinction between established input and bridge extension is essential. The S^3 harmonic spectrum and cubic-Galileon background are treated as the cosmological starting point. The Schur/Feshbach reduction is an exact mathematical operation once a physical complement is specified. The identification of its low-energy coefficients with the topographic response is a bridge assumption to be tested, not a declaration that the complete microscopic BHSM action has already been reduced to four-dimensional cosmology.

The paper makes four main contributions. First, it standardizes the S^3 harmonic convention and shows how the observed sky dipole arises from the four-dimensional $n = 1$ eigenspace rather than by identifying a radial mode with an angular dipole. Second, it derives a reduced kernel in which a positive higher-spatial-gradient term can emerge after integrating out a stable complement. Third, it identifies a parameter interval in which the $n = 1$ response is selectively softened while all $n \geq 2$ harmonics are stiffened. Fourth, it gives an executable preregistration protocol for curvature, supernova anisotropy, growth, and lensing tests.

The claim boundary is therefore:

This work does not assume that the scalar-topographic cosmological field has already been derived from the complete microscopic Berger–Hopf action. Exact reductions are distinguished from cross-scale identifications and exploratory reference-branch assumptions. Prospective observables are frozen before comparison with designated independent data.

2 Hyperspherical Parent and Harmonic Convention

We use a closed FLRW geometry

$$ds^2 = -dt^2 + a^2(t)R_H^2 \left[d\psi^2 + \sin^2 \psi d\Omega_2^2 \right],$$

with $a_0 = 1$. Define

$$D_C(z) = c \int_0^z \frac{dz'}{H(z')}, \quad \psi(z) = \frac{D_C(z)}{R_H}.$$

Scalar harmonics satisfy

$$-\nabla_\gamma^2 Y_n = \frac{n(n+2)}{R_H^2} Y_n, \quad n = 0, 1, 2, \dots,$$

so the physical spatial eigenvalue is

$$p_n(a) = \frac{n(n+2)}{a^2 R_H^2}.$$

A common closed-universe convention uses $k^2 = N^2 - 1$, hence

$$\boxed{N = n + 1}.$$

2.1 Exceptional first nonconstant harmonic

For our $n = 1$, $k^2 = 3$ and $N = 2$. The degree-one scalar harmonics on the unit three-sphere obey

$$D_i D_j Q^{(1)} = -\gamma_{ij} Q^{(1)}.$$

Therefore

$$\left(D_i D_j + \frac{k^2}{3} \gamma_{ij} \right) Q^{(1)} = 0.$$

The scalar-derived trace-free metric harmonic is absent. For the regular generally covariant one-scalar sector, the remaining scalar gauge transformations and lapse/shift constraints leave no ordinary propagating scalar degree of freedom in this exceptional sector. Modern gauge-invariant closed-universe analyses likewise take the propagating scalar spectrum to begin at alternative-convention $N \geq 3$ [2, 3].

Accordingly, the minimal bridge no longer uses our $n = 1$ as the physical scalar-metric carrier. An additional boundary degree of freedom could evade this conclusion, but it is not required for the construction below.

2.2 First physical scalar harmonic and exact dipole subspace

The first propagating scalar harmonic is

$$\boxed{n_\star = 2}, \quad \boxed{N_\star = 3}, \quad p_2 = \frac{8}{a^2 R_H^2}.$$

Let the unit $S^3 \subset \mathbb{R}^4$ embedding coordinates be

$$X^A = (\cos \psi, \sin \psi \hat{n}^i).$$

Every real degree-two scalar harmonic can be written

$$T_2 = A_{AB} X^A X^B, \quad A_{AB} = A_{BA}, \quad \delta^{AB} A_{AB} = 0.$$

Writing $c = \cos \psi$, $s = \sin \psi$, $a_i = A_{4i}$, and

$$A_{ij} = S_{ij} - \frac{A_{44}}{3} \delta_{ij}, \quad S_i^i = 0,$$

gives the exact observer-shell decomposition

$$\boxed{T_2 = A_{44} \left(c^2 - \frac{s^2}{3} \right) + 2cs \mathbf{a} \cdot \hat{\mathbf{n}} + s^2 S_{ij} \hat{n}^i \hat{n}^j.}$$

These terms are respectively a sky monopole, dipole, and quadrupole. The pure cross-term subspace

$$A_{44} = 0, \quad S_{ij} = 0, \quad a_i \neq 0$$

therefore gives an exact sky dipole:

$$\boxed{T_{2,\text{dip}} = A \sin(2\psi) (\hat{\mathbf{n}} \cdot \hat{\mathbf{p}}).}$$

Thus the gauge audit shifts the carrier to the first physical compact-space scalar mode while retaining an exact dipolar observer-sky sector.

3 Constraint-Reduced Response and the Topographic Kernel

Let $\hat{\mathcal{H}}$ be the physical Hessian after constraints and gauge directions have been removed. For a selected topographic branch P and physical complement Q , the exact Schur/Feshbach reduced operator is

$$\mathcal{K} = P \left[\hat{\mathcal{H}} - \hat{\mathcal{H}} Q (Q \hat{\mathcal{H}} Q)^{-1} Q \hat{\mathcal{H}} \right] P.$$

For one scalar harmonic, write the quadratic block as

$$\hat{\mathcal{H}}_n = \begin{pmatrix} A(p) & C(p) \\ C^\dagger(p) & D(p) \end{pmatrix}.$$

The exact reduced stiffness is

$$K_{\text{red}}(p) = A(p) - C(p)D^{-1}(p)C^\dagger(p).$$

A minimal long-wavelength realization is

$$A(p) = Z_0 p, \quad D(p) = M_c^2 + c_c p, \quad C(p) = g\sqrt{p}.$$

Then

$$K_{\text{red}}(p) = Z_0 p - \frac{g^2 p}{M_c^2 + c_c p}.$$

For $c_c p \ll M_c^2$,

$$K_{\text{red}}(p) = \left(Z_0 - \frac{g^2}{M_c^2} \right) p + \frac{g^2 c_c}{M_c^4} p^2 + O(p^3).$$

Thus a positive p^2 stiffness can arise from a coupled second-order parent system after a stable complement is integrated out. In this interpretation the old static operator

$$-\nabla^2 + B\nabla^4$$

is a low-energy spatial derivative expansion, not evidence for a fundamental fourth-order time equation.

Define

$$\chi \equiv \frac{g^2}{M_c^2}, \quad L_c^2 \equiv \frac{c_c}{M_c^2}.$$

After canonical normalization of the leading spatial term, the reduced kernel may be written schematically as

$$K_{\text{topo}}(p) = m_T^2 + (c_s^2 - \chi)p + \chi L_c^2 p^2 + \dots.$$

The signs of the temporal kinetic term and of the full physical gradient matrix must be checked independently; a positive spatial p^2 coefficient by itself is not a complete ghost analysis.

4 Cubic-Galileon Scalar–Metric Bridge

The cosmological parent action is taken to be

$$S = \int d^4x \sqrt{-g} \left[\frac{M_{\text{Pl}}^2}{2} R - \frac{1}{2} \nabla_\mu T \nabla^\mu T + \frac{1}{\Lambda^3} \square T (\nabla T)^2 - V(T) + \mathcal{L}_m \right].$$

Using

$$X \equiv -\frac{1}{2} \nabla_\mu T \nabla^\mu T,$$

this corresponds to

$$G_2 = X - V(T), \quad G_3 = \frac{2X}{\Lambda^3}, \quad G_4 = \frac{M_{\text{Pl}}^2}{2}, \quad G_5 = 0$$

in the convention $\mathcal{L}_3 = -G_3 \square T$.

The model belongs to the kinetic-gravity-braiding subclass of Horndeski theory [4, 5]. Since G_4 is constant and $G_5 = 0$,

$$M_*^2 = M_{\text{Pl}}^2, \quad \alpha_M = 0, \quad \alpha_T = 0.$$

The scalar–metric mixing is controlled by the braiding function

$$\alpha_B = \frac{2\dot{T}^3}{HM_{\text{Pl}}^2\Lambda^3},$$

and

$$\alpha_K = \frac{\dot{T}^2}{H^2 M_{\text{Pl}}^2} + 6\alpha_B.$$

In the Bellini–Sawicki organization of Horndeski perturbations the scalar kinetic combination is

$$D_{\text{kin}} = \alpha_K + \frac{3}{2}\alpha_B^2.$$

Their perturbation equations are formulated for a spatially flat background [5]. We therefore use $D_{\text{kin}} > 0$ below as an inherited kinetic diagnostic for the cubic-Galileon sector, not as a complete proof of perturbative stability for the present closed background plus the additional reduced spatial response. A full stability proof requires the constrained gauge-invariant kinetic and gradient matrices, including the exceptional lowest scalar sector identified above.

For numerical work we softly break shift symmetry with

$$V(T) = V_0 - \mu^3 T.$$

The scalar energy density and pressure are

$$\rho_T = \frac{1}{2}\dot{T}^2 + V(T) + \frac{6H\dot{T}^3}{\Lambda^3},$$

$$P_T = \frac{1}{2}\dot{T}^2 - V(T) - \frac{2\dot{T}^2\ddot{T}}{\Lambda^3}.$$

The shift current

$$J = \dot{T} + \frac{6H\dot{T}^2}{\Lambda^3}$$

obeys

$$\dot{J} + 3HJ = \mu^3.$$

These background expressions are the standard kinetic-gravity-braiding relations specialized to the present G_2, G_3 choice [4].

The closed-FLRW background equation is

$$3M_{\text{Pl}}^2 \left(H^2 + \frac{1}{a^2 R_H^2} \right) = \rho_m + \rho_r + \rho_T.$$

The bridge does not require the bare Horndeski sound speed to become negative. Instead, the BHSM-inspired reduced response is used to lower the effective long-wavelength spatial stiffness while a positive p^2 term stiffens shorter wavelengths. This statement concerns the reduced spatial kernel; it is not, by itself, a complete closed-FLRW stability theorem.

5 BHSM-native transport of the physical topographic mode

The cosmological bridge is formulated first in the mode ontology of the Berger–Hopf Standard Model (BHSM), and only secondarily in a four-dimensional effective-field-theory coordinate system. A topographic carrier is therefore not introduced as an additional free bulk scalar. It is an action-owned geometric/boundary mode whose physical dynamics are assigned after the constraint and gauge quotients have been taken.

5.1 Physical tangent space and reduced operator

At a background configuration $z_\star$, define

$$T_{\text{phys}}(z_\star) = \ker DC(z_\star) \cap \mathcal{G}^\perp, \quad (1)$$

where $\mathcal{C}$ is the action-owned constraint map and $\mathcal{G}$ is the space of infinitesimal gauge directions. Let

$$\hat{\mathcal{H}} = P_{\text{phys}} D^2 \mathcal{L}_{\text{KKT}}(z_\star) P_{\text{phys}} \quad (2)$$

be the physical Hessian. For the first admissible scalar-topographic harmonic projector P_2 , define the reduced operator

$$\mathcal{K}_2 = P_2 \left[\hat{\mathcal{H}} - \hat{\mathcal{H}} Q_2 \left(Q_2 \hat{\mathcal{H}} Q_2 \right)^{-1} Q_2 \hat{\mathcal{H}} \right] P_2, \quad (3)$$

where $Q_2 = 1 - P_2$ on the already physical tangent space and the inverse is taken only on the admissible eliminated block.

5.2 Action-owned $n = 2$ envelope transport theorem

Theorem 1. Assume that the cosmological parent envelope is $\mathcal{E}_{\text{cos}} \simeq S^3(R_H)$, that scalar/topographic perturbations are reduced to T_{phys} before physical propagation is assigned, that $n = 0$ is absorbed into the homogeneous background, and that the exceptional regular closed-FLRW scalar $n = 1$ sector is removed by the physical quotient. If the retained $n = 2$ kinetic form is positive and nondegenerate, then the first physical scalar-topographic carrier is an action-owned $n = 2$ envelope mode. Its canonical momentum is generated by the same reduced action and is not a separately postulated bulk field.

Proof. Let $e_2(x)$ be a normalized element of $P_2 T_{\text{phys}}$ and write

$$\delta \Phi_{\text{topo}}(t, x) = q_2(t) e_2(x). \quad (4)$$

The gauge- and constraint-reduced quadratic action has the one-mode form

$$S_2^{(2)} = \frac{1}{2} \int dt a^3 \left[\mathcal{G}_2(t) (D_t q_2)^2 - \mathcal{M}_2^2(t) q_2^2 + 2 \mathcal{J}_2(t) q_2 \right], \quad (5)$$

with coefficients obtained from matrix elements of the reduced physical operator. The conjugate momentum is

$$\Pi_2 = a^3 \mathcal{G}_2 D_t q_2, \quad (6)$$

and variation gives

$$D_t \left(a^3 \mathcal{G}_2 D_t q_2 \right) + a^3 \mathcal{M}_2^2 q_2 = a^3 \mathcal{J}_2. \quad (7)$$

Thus (q_2, Π_2) is the reduced phase-space pair of one action-owned geometric mode. $\square$

5.3 Effective closed-Horndeski matching

The exact closed-Horndeski $n = 2$ result derived elsewhere in this manuscript is used as an effective kinetic representation of the same retained mode, rather than as its microscopic ontology. In the Bellini–Sawicki convention,

$$\mathcal{G}_{S,2} = M_{\text{Pl}}^2 \frac{5D_{\text{kin}}}{D_{\text{kin}} + 10(1 - \alpha_B^{\text{BS}}/2)^2}. \quad (8)$$

Hence $D_{\text{kin}} > 0$ and a nonsingular denominator provide the effective positive-kinetic matching condition used on the reference branch. The EFT Stueckelberg coordinate π is consequently interpreted as a coordinate representative of the retained response of q_2 , not as an additional BHSM degree of freedom.

5.4 Exact observer dipole

A degree-two scalar harmonic on S^3 can be written

$$T_2 = A_{AB} X^A X^B, \quad A_A{}^A = 0. \quad (9)$$

On an observer shell,

$$T_2 = A_{44} \left(c^2 - \frac{s^2}{3} \right) + 2cs \mathbf{a} \cdot \hat{\mathbf{n}} + s^2 S_{ij} \hat{n}^i \hat{n}^j. \quad (10)$$

The pure A_{4i} subspace therefore gives the exact sky dipole

$$\boxed{T_{2,\text{dip}} = A \sin(2\psi) (\hat{\mathbf{n}} \cdot \hat{\mathbf{p}})}. \quad (11)$$

5.5 Curved-EFT compatibility status

The curvature-aware EFT system remains an important representation-level cross-check. It is not used to define the existence or mode count of the BHSM carrier. The literal curved constraint/evolution equations and the reference background have been transcription-audited, while the global curved-EFT differential-algebraic implementation retains a nonzero constraint-preservation defect. That unresolved object is therefore classified as an EFT compatibility audit rather than as the foundational transport law. The defining bridge evolution is Eq. (7).

6 Exact Closed-Horndeski Kernel for the First Physical Scalar Mode

The previous section uses the Bellini–Sawicki functions to organize the cubic-Galileon background. For the curvature-scale mode, however, the flat-FLRW perturbation formulas are not sufficient. We therefore specialize the non-flat Horndeski quadratic action derived by Akama and Kobayashi [6].

For

$$G_2 = X - V(T), \quad G_3 = \frac{2X}{\Lambda^3}, \quad G_4 = \frac{M_{\text{Pl}}^2}{2}, \quad G_5 = 0,$$

their tensor coefficients reduce exactly to

$$\mathcal{F}_T = \mathcal{G}_T = M_{\text{Pl}}^2.$$

Because G_4 is constant and $G_5 = 0$, the explicit curvature contributions to their Θ and Σ vanish. The remaining coefficients are

$$\Theta = HM_{\text{Pl}}^2 - \frac{\dot{T}^3}{\Lambda^3}$$

and

$$\Sigma = \frac{1}{2}\dot{T}^2 + \frac{12H\dot{T}^3}{\Lambda^3} - 3M_{\text{Pl}}^2H^2.$$

Equivalently,

$$\Theta = HM_{\text{Pl}}^2 \left(1 - \frac{\alpha_B}{2}\right).$$

Define

$$r(t) \equiv \frac{M_{\text{Pl}}^2 \Sigma}{\Theta^2}, \quad \kappa(t) \equiv \frac{K}{a^2}.$$

The first physical scalar harmonic has alternative-convention index $N = 3$, so

$$k^2 = K(N^2 - 1) = 8K.$$

The two Akama–Kobayashi scalar constraints then reduce to

$$\Sigma \delta n + 8\Theta \kappa \chi + 3\Theta \dot{\zeta} + 5M_{\text{Pl}}^2 \kappa \zeta = 0,$$

$$\Theta \delta n - M_{\text{Pl}}^2 \dot{\zeta} - M_{\text{Pl}}^2 \kappa \chi = 0.$$

They can be solved algebraically:

$$\delta n = \frac{5M_{\text{Pl}}^2}{\Theta(r+8)} \left[\dot{\zeta} - \frac{M_{\text{Pl}}^2 \kappa}{\Theta} \zeta \right]$$

and

$$\chi = -\frac{r+3}{\kappa(r+8)} \dot{\zeta} - \frac{5M_{\text{Pl}}^2}{\Theta(r+8)} \zeta.$$

The reduced kinetic coefficient of this $n = 2$ mode is therefore

$$\mathcal{G}_{S,2} = M_{\text{Pl}}^2 \frac{5(r+3)}{r+8}.$$

Using the cubic-Galileon α -functions gives the exact identity

$$r+3 = \frac{D_{\text{kin}}}{2(1 - \alpha_B/2)^2},$$

so that

$$\mathcal{G}_{S,2} = M_{\text{Pl}}^2 \frac{5D_{\text{kin}}}{D_{\text{kin}} + 10(1 - \alpha_B/2)^2}.$$

Thus, provided $\Theta \neq 0$, positivity of the previously computed D_{kin} is sufficient for positivity of the pure gravity–scalar kinetic coefficient of the first physical closed-universe scalar harmonic.

For completeness, the corresponding non-flat Horndeski gradient coefficient is

$$\mathcal{F}_{S,2} = \frac{1}{a} \frac{d}{dt} \left[\frac{5}{r+8} \left(\frac{aM_{\text{Pl}}^4}{\Theta} \right) \right] - M_{\text{Pl}}^2 + \frac{5}{r+8} \frac{M_{\text{Pl}}^6}{\Theta^2} \frac{K}{a^2}.$$

The repository evaluates this expression as a gravity–scalar diagnostic only. The late-time R1 background contains matter and radiation, whose perturbations introduce additional dynamical variables and source terms. Consequently, $\mathcal{F}_{S,2}$ is not promoted here to a complete matter-era stability or light-cone transfer result.

This section closes an important part of the bridge: the first physical harmonic has an explicit constrained scalar–metric kernel rather than only a geometric identification. The remaining observable calculation is the coupled matter/radiation plus topographic-response system and its gauge-invariant transformation to the metric and velocity variables entering distance, growth, and lensing observables.

7 Matter-Sourced Constraint Transfer at n=2

The Round-3 kernel established the pure gravity–scalar constrained dynamics of the first physical closed-universe harmonic. The next step is to introduce minimally coupled matter sources without importing a flat-space or quasi-static approximation. We use the curvature-aware Newtonian-gauge EFT equations written by Gambino and Pace [7]. Their metric convention is

$$ds^2 = -(1 + 2\Phi)dt^2 + a^2(1 - 2\Psi)\gamma_{ij}dx^i dx^j,$$

with

$$\delta T^0_0 = -\delta\rho, \quad \delta T^0_i = (\rho + p)D_i v.$$

For the present cubic-Galileon subclass,

$$\alpha_M = \alpha_T = \alpha_H = 0, \quad M^2 = M_{\text{Pl}}^2.$$

For ideal dust and ideal radiation we set the scalar anisotropic stress to zero, so the traceless spatial equation gives

$$\boxed{\Phi = \Psi}.$$

On the physical $n = 2$ harmonic,

$$D^2 Q = -8KQ, \quad (D^2 + 3K)Q = -5KQ.$$

Writing

$$\kappa = \frac{K}{a^2},$$

the sourced 00 equation becomes

$$\begin{aligned} & 6(1 + \alpha_B)H\dot{\Phi} + (\alpha_K - 6\alpha_B)H^2\dot{\pi} + \left[(6 - \alpha_K + 12\alpha_B)H^2 + 10\kappa\right]\Phi \\ & + 6H\left[\frac{\rho + p}{2M_{\text{Pl}}^2} + \dot{H}(1 + \alpha_B) + \kappa\left(1 - \frac{8}{3}\alpha_B\right)\right]\pi = -\frac{\delta\rho}{M_{\text{Pl}}^2}. \end{aligned}$$

The $0i$ equation is

$$2\dot{\Phi} - 2H\alpha_B\dot{\pi} + 2(1 + \alpha_B)H\Phi + \left(2\dot{H} + \frac{\rho + p}{M_{\text{Pl}}^2}\right)\pi = -\frac{q}{M_{\text{Pl}}^2},$$

where

$$q = \rho_m v_m + \frac{4}{3}\rho_r v_r, \quad \delta\rho = \rho_m \delta_m + \rho_r \delta_r.$$

These equations form a 2×2 local constraint-transfer system for

$$(\dot{\Phi}, \dot{\pi}).$$

Its determinant is

$$\det \mathcal{C} = -2H^2 \left(\alpha_K + 6\alpha_B^2 \right) = -2H^2 \left(D_{\text{kin}} + \frac{9}{2}\alpha_B^2 \right).$$

Thus the same positive- D_{kin} branch used in the background and Round-3 kinetic audit guarantees that this sourced rate block is nonsingular.

For the velocity convention above, covariant conservation gives

$$\dot{\delta}_m = \frac{k^2}{a^2} v_m + 3\dot{\Phi}, \quad \dot{v}_m = -\Phi,$$

and

$$\dot{\delta}_r = \frac{4}{3} \frac{k^2}{a^2} v_r + 4\dot{\Phi}, \quad \dot{v}_r = H v_r - \Phi - \frac{1}{4} \delta_r,$$

with

$$\frac{k^2}{a^2} = 8\kappa.$$

The repository implements these equations as a local linear transfer matrix with the additional topographic response explicitly switched off. This is deliberately not yet promoted to a globally integrated observable solution. The curvature-aware EFT equations and the Round-3 ADM kernel use different perturbation variables and normalizations; the remaining evolution equations must first be matched analytically and then subjected to a Bianchi/constraint-preservation audit. Only after that compatibility gate is passed will we integrate the full matter-era $n = 2$ system and add the independent reduced topographic operator.

8 Selective Harmonic Response

Let

$$L_c(a) = a R_H \ell, \quad L_c^2 p_n = \ell^2 n(n+2),$$

and write

$$K_n = p_n C_n, \quad C_n = c_s^2 - \chi \left[1 - \ell^2 n(n+2) \right].$$

The physical scalar spectrum begins at $n = 2$. Selective softening of $n = 2$ with stiffening of every propagating $n \geq 3$ requires

$$1 - 8\ell^2 > 0, \quad 1 - 15\ell^2 < 0,$$

so

$$\frac{1}{15} < \ell^2 < \frac{1}{8}.$$

Then

$$C_2 < c_s^2, \quad C_n > c_s^2 \quad (n \geq 3).$$

The formal $n = 1$ coefficient also lies on the soft side, but $n = 1$ is not a propagating scalar degree of freedom in the regular one-field closed-FLRW sector.

If $0 < C_2 \ll 1$,

$$T_2 \simeq \frac{S_2}{p_2 C_2}$$

can be enhanced while higher physical harmonics remain regular. For the next physical level,

$$\frac{T_3}{T_2} = \frac{S_3}{S_2} \frac{8C_2}{15C_3},$$

so

$$\left| \frac{T_3/S_3}{T_2/S_2} \right| = \frac{8C_2}{15C_3} < \frac{8}{15}.$$

For the pure-dipole A_{4i} subspace of the selected $n = 2$ harmonic, the observer-sky pattern is exactly dipolar with geometric radial envelope $\sin 2\psi$.

9 Observable Projection

The observable distance perturbation is not identified directly with the scalar amplitude. At linear order it is a light-cone projection of metric, velocity, and scalar perturbations:

$$\frac{\delta D_L}{\bar{D}_L} = \int d\chi \left[W_\Phi \Phi + W_\Psi \Psi + W_v v_\parallel + \dots \right].$$

If the selected $n = 2$ curvature-scale sector admits a physical gauge-invariant metric response, its exact scalar angular dependence can factor as

$$\hat{\mathbf{n}} \cdot \hat{\mathbf{p}},$$

leading schematically to

$$\frac{\delta D_L}{\bar{D}_L} = \epsilon_D(z) \hat{\mathbf{n}} \cdot \hat{\mathbf{p}}.$$

The existence and normalization of $\epsilon_D(z)$ are therefore bridge outputs to be derived, not consequences of the scalar harmonic geometry alone.

For supernovae,

$$\mu = 5 \log_{10} D_L + \text{const},$$

and therefore a small magnitude dipole A_μ corresponds to

$$\frac{\delta D_L}{D_L} = \frac{\ln 10}{5} A_\mu \cos \theta.$$

For structure growth we define

$$k^2 \Psi = -4\pi G a^2 \mu(k, a) \rho_m \delta_m, \quad \Phi = \gamma(k, a) \Psi,$$

and

$$\Sigma(k, a) = \frac{\mu(k, a)}{2} [1 + \gamma(k, a)].$$

The matter growth factor obeys

$$D'' + \left(2 + \frac{H'}{H} \right) D' - \frac{3}{2} \Omega_m(a) \mu(k, a) D = 0,$$

with prime denoting $d/d \ln a$. The observable is

$$f\sigma_8(k, z) = \frac{D'}{D} D \sigma_{8,\text{ini}} = D'(k, z) \sigma_{8,\text{ini}}.$$

The quasi-static limit is used only on scales inside the scalar sound horizon [8]. Horizon-scale $n = 1$ observables require the full constrained long-wavelength response and must not be obtained by extrapolating quasi-static formulas into the first physical curvature-scale mode.

10 Observable transfer theorem for the physical $n = 2$ mode

The reduced BHSM mode becomes observationally useful only after its action-generated phase-space data are mapped to metric and matter response variables. This section defines that map without introducing an additional degree of freedom.

Let

$$X_2(t) = \begin{pmatrix} q_2(t) \\ \Pi_2(t) \end{pmatrix} \quad (12)$$

denote the physical $n = 2$ phase-space state. Linear response of the retained metric/matter variables can be represented abstractly as

$$Y_2(t) = \mathbb{T}_2(t, t_i) X_2(t_i) + \int_{t_i}^t \mathbb{G}_2(t, t') S_2(t') dt', \quad (13)$$

where Y_2 may contain the two scalar metric potentials, matter density and velocity response, and any auxiliary response coordinate used by the chosen four-dimensional representation. The operators $\mathbb{T}_2$ and $\mathbb{G}_2$ are determined by the reduced action and its source projection; they are not independent phenomenological fields.

10.1 Angular-factorization theorem

Theorem 2 (single-mode observable factorization). Assume (i) the retained perturbation is the pure A_{4i} component of the physical $n = 2$ scalar-topographic mode, (ii) the response is linear, and (iii) the transfer operator is rotationally covariant on the closed background. Then every scalar linear observable $\mathcal{O}$ sourced by this single mode has the leading angular form

$$\boxed{\delta\mathcal{O}(z, \hat{\mathbf{n}}) = \mathcal{A}_{\mathcal{O}}(z) (\hat{\mathbf{n}} \cdot \hat{\mathbf{p}})}, \quad (14)$$

where the entire observable dependence is contained in the redshift transfer amplitude $\mathcal{A}_{\mathcal{O}}(z)$ and the sky axis $\hat{\mathbf{p}}$ is shared by all observables at leading one-mode order.

Proof. The physical perturbation belongs to one $n = 2$ representation of the isometry group of S^3 . A rotationally covariant linear transfer operator cannot generate a different harmonic representation in the absence of an additional anisotropic source. Restriction of the retained A_{4i} subspace to the observer shell is exactly $T_{2,\text{dip}} \propto \hat{\mathbf{n}} \cdot \hat{\mathbf{p}}$ by Eq. (11). Applying any scalar linear observation functional changes only its radial/time coefficient. Therefore the angular factor is common and Eq. (14) follows. $\square$

10.2 Observable kernels

For each observable define a linear functional $\mathcal{L}_{\mathcal{O},z}$ acting on Eq. (13). Then

$$\mathcal{A}_{\mathcal{O}}(z) = \mathcal{L}_{\mathcal{O},z} \left[\mathbb{T}_2 X_2(t_i) + \int \mathbb{G}_2 S_2 dt' \right]. \quad (15)$$

The principal late-time observables used in this work are represented by

$$\frac{\delta D_L}{D_L} = \mathcal{A}_{D_L}(z) (\hat{\mathbf{n}} \cdot \hat{\mathbf{p}}), \quad (16)$$

$$\frac{\delta H}{H} = \mathcal{A}_H(z) (\hat{\mathbf{n}} \cdot \hat{\mathbf{p}}), \quad (17)$$

$$\frac{\delta D_A}{D_A} = \mathcal{A}_{D_A}(z) (\hat{\mathbf{n}} \cdot \hat{\mathbf{p}}), \quad (18)$$

$$\delta(\Phi + \Psi) = \mathcal{A}_{\text{lens}}(z) (\hat{\mathbf{n}} \cdot \hat{\mathbf{p}}), \quad (19)$$

$$\frac{\delta(f\sigma_8)}{f\sigma_8} = \mathcal{A}_{f\sigma_8}(z) (\hat{\mathbf{n}} \cdot \hat{\mathbf{p}}), \quad (20)$$

whenever the corresponding observable is evaluated in the linear response regime.

For supernova distance modulus,

$$\delta\mu = \frac{5}{\ln 10} \frac{\delta D_L}{D_L}, \quad (21)$$

so that

$$\boxed{A_\mu(z) = \frac{5}{\ln 10} \mathcal{A}_{D_L}(z)}. \quad (22)$$

Equation (22) supplies the direct bridge between the geometric $n = 2$ carrier and the low-redshift supernova dipole amplitude.

10.3 Shared-axis and amplitude-ratio predictions

Once the transfer kernels are fixed by the reduced action, the amplitudes are not independent fit parameters. For any two linear observables,

$$\frac{\mathcal{A}_{\mathcal{O}_1}(z)}{\mathcal{A}_{\mathcal{O}_2}(z)} = \frac{\mathcal{L}_{\mathcal{O}_1,z}[\mathcal{R}_2]}{\mathcal{L}_{\mathcal{O}_2,z}[\mathcal{R}_2]}, \quad (23)$$

where $\mathcal{R}_2$ denotes the common physical response generated by X_2 and its projected sources. Thus the single-mode branch predicts a common leading axis and linked redshift-dependent amplitude ratios across distance, expansion, growth, and lensing observables.

A statistically persistent axis mismatch between independent observables, after survey-window and calibration effects are controlled, would falsify the pure one-mode realization. Likewise, a leading angular morphology that cannot be represented by the A_{4i} dipole subspace requires additional physical harmonics or rejection of the one-mode branch.

10.4 Higher-mode control

For the harmonic filter

$$C_n = c_s^2 - \chi \left[1 - \ell^2 n(n+2) \right], \quad (24)$$

the interval

$$\frac{1}{15} < \ell^2 < \frac{1}{8} \quad (25)$$

softens the physical $n = 2$ carrier while stiffening every $n \geq 3$ scalar harmonic. The next source-normalized response obeys

$$\left| \frac{T_3/S_3}{T_2/S_2} \right| = \frac{8C_2}{15C_3}. \quad (26)$$

This provides the explicit control parameter for departures from the single-axis theorem.

10.5 Closure criterion for the manuscript

Scientific closure of the observable bridge requires the reduced operator $\mathcal{K}_2$ to determine the transfer kernels entering Eq. (15). A four-dimensional EFT realization is acceptable only if it reproduces the same two-dimensional reduced phase space (q_2, Π_2) and the same observable transfer kernels without adding an independent physical scalar. Reference-branch numerical curves may be quoted before this exact identification only when they are explicitly labeled as reference predictions rather than framework-wide theorems.

11 Predictions and Falsification Criteria

11.1 Low-redshift calibration

The Pantheon+ data release and cosmological analysis provide the supernova basis for the carried-forward tomographic calibration used here [9, 10]. The earlier scalar-topographic analysis reported

$$A_\mu^\star = -0.0412 \pm 0.009$$

in the bin

$$0.01 < z < 0.03,$$

with best-fit direction

$$(\text{RA}, \text{Dec}) = (211.48^\circ, -12.81^\circ).$$

This corresponds to

$$\frac{\delta D_L}{D_L} = (-0.01895 \pm 0.00414) \cos \theta,$$

and at sufficiently low redshift,

$$\frac{\delta H}{H} = (+0.01895 \pm 0.00414) \cos \theta.$$

These numbers are a carried-forward calibration input; the present repository does not independently rerun the Pantheon+ tomographic likelihood. They normalize only the observed $n = 1$ response,

$$A_\mu = \frac{\mathcal{K}_{\text{SN}} S_2}{p_2 C_2},$$

and do not independently determine S_2 , $\mathcal{K}_{\text{SN}}$, or C_2 . Their interpretation as a metric/light-cone response is additionally conditional on resolving the exceptional closed-FLRW $n = 1$ sector.

11.2 Curvature

The pre-existing prospective curvature forecast is retained without retuning:

$$\boxed{\Omega_k = -0.018 \pm 0.004}.$$

At $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$,

$$R_H = \frac{c}{H_0 \sqrt{|\Omega_k|}} \simeq 33.15 \text{ Gpc}.$$

The sign prediction is $\Omega_k < 0$ in the usual FLRW curvature convention.

This value is a frozen model forecast, not a current empirical best fit. DESI DR2 BAO results are reported as well described by flat Λ CDM, even while combinations with other probes motivate tests of extensions such as dynamical dark energy [11]. The purpose of retaining -0.018 ± 0.004 is therefore prospective falsification: it must be compared against independent curvature inference without being moved toward the observed result.

11.3 Harmonic hierarchy

The bridge predicts selective $n = 2$ softening with no analogous $n \geq 3$ enhancement. A future source-normalized S^3 $n = 2$ susceptibility comparable to or larger than the $n = 1$ susceptibility would reject the minimal harmonic-filter branch. This statement is made at the compact-space harmonic level; comparison to a measured sky quadrupole requires an explicit observer-shell projection.

11.4 Fixed-axis test

A prospective independent supernova sample should first test the carried-forward axis

$$(211.48^\circ, -12.81^\circ)$$

rather than scanning freely over the sky. A free-axis search may be reported secondarily.

11.5 High- n recovery

At large n ,

$$C_n \simeq \chi \ell^2 n^2,$$

so the extra scalar susceptibility falls as n^{-2} . The model therefore predicts a strong separation between the first nonconstant compact-space level and conventional clustering modes.

11.6 Failure conditions

The minimal bridge is disfavored if one or more of the following are established with adequate precision:

1. independent curvature measurements exclude the frozen negative- Ω_k forecast;
2. the source-normalized S^3 $n = 2$ response is comparable to or exceeds $n = 1$;
3. an independent low-redshift sample excludes the frozen dipole axis while finding a stable incompatible physical axis;

4. a coherent metric/light-cone dipole of comparable amplitude persists into a redshift regime in which the completed model predicts loss of enhanced susceptibility;
5. ordinary RSD modes show a large scale-independent modification incompatible with the high- n suppression;
6. lensing requires a metric response incompatible with the same completed scalar–metric propagator that controls matter clustering;
7. no gauge-invariant physical metric response exists for the exceptional closed-FLRW harmonic corresponding to the proposed $n = 1$ topographic mode.

12 Numerical Implementation and Preregistration Protocol

The numerical pipeline is separated into

$$\boxed{\text{background} \rightarrow \text{physical scalar response} \rightarrow \text{observables} \rightarrow \text{comparison.}}$$

Prediction generation and data comparison are implemented in separate scripts.

12.1 Background system

Let $v = \dot{T}$. Expanding the scalar current equation gives

$$\left(1 + \frac{12Hv}{\Lambda^3}\right) \dot{v} + \frac{6v^2}{\Lambda^3} \dot{H} = \mu^3 - 3H \left(v + \frac{6Hv^2}{\Lambda^3}\right).$$

The Raychaudhuri equation gives

$$\frac{2v^2}{\Lambda^3} \dot{v} - 2M_{\text{Pl}}^2 \dot{H} = \rho_m + \frac{4}{3}\rho_r + v^2 + \frac{6Hv^3}{\Lambda^3} - \frac{2M_{\text{Pl}}^2}{a^2 R_H^2}.$$

At each integration step these equations form a 2×2 linear system for $(\dot{v}, \dot{H})$. The remaining equations are

$$\dot{T} = v, \quad \dot{a} = aH.$$

The Friedmann equation is retained as an independent residual check:

$$3M_{\text{Pl}}^2 \left(H^2 + \frac{1}{a^2 R_H^2} \right) - (\rho_m + \rho_r + \rho_T) = 0.$$

12.2 Dimensionless variables

We integrate in $N = \ln a$ with

$$E = \frac{H}{H_0}, \quad \phi = \frac{T}{M_{\text{Pl}}}, \quad \lambda = \frac{\Lambda^3}{M_{\text{Pl}} H_0^2}, \quad q = \frac{\mu^3}{M_{\text{Pl}} H_0^2}.$$

The exploratory reference branch R1 uses

$$\Omega_{m0} = 0.31, \quad \Omega_{r0} = 9.0 \times 10^{-5}, \quad \Omega_{k0} = -0.018, \quad \lambda = 1, \quad q = 3.$$

The constant V_0 is determined only by the shooting condition $E(a = 1) = 1$.

12.3 Initial conditions

The numerical integration begins at

$$z_i = 100.$$

The decaying homogeneous current solution is removed using the matter-era particular solution

$$J_i \simeq \frac{2\mu^3}{9H_i}.$$

The initial scalar velocity is the positive small-velocity root of

$$v_i + \frac{6H_i v_i^2}{\Lambda^3} = J_i.$$

12.4 Growth grid

Before prospective comparison, predictions are generated on the fixed grid

$$k = \{0.02, 0.05, 0.10, 0.15, 0.20\} \text{ Mpc}^{-1}$$

and

$$z = \{0.5, 0.8, 1.0, 1.5, 2.1\}.$$

A deterministic JSON manifest containing the action version, parameters, numerical tolerances, calibration values, and predicted observables is hashed with SHA-256. Any later scientific or numerical revision creates a new manifest version rather than overwriting the previous freeze.

13 Results: Reference Branch R1

R1 is an exploratory reference branch rather than a uniquely derived BHSM cosmology. Its purpose is to demonstrate that the bridge can simultaneously realize late-time scalar activation, a selectively enhanced $n = 1$ response, and near-standard high- n structure growth.

For

$$\Omega_{m0} = 0.31, \quad \Omega_{r0} = 9.0 \times 10^{-5}, \quad \Omega_{k0} = -0.018, \quad \lambda = 1, \quad q = 3,$$

the shooting condition $E(1) = 1$ gives

$$\boxed{\frac{V_0}{M_{\text{Pl}}^2 H_0^2} = 2.39861.}$$

The scalar braiding and no-ghost combination evolve as shown in Table 1.

The scalar equation of state evolves from nearly vacuum-like behavior at high redshift to

$$w_T(z = 0) \simeq -0.906,$$

and the background acceleration transition occurs at

$$z_{\text{acc}} \simeq 0.683.$$

Relative to a curved Λ CDM model with identical $H_0, \Omega_{m0}, \Omega_{r0}, \Omega_{k0}$, the R1 expansion rate differs by only order one percent over the late-time interval. The maximum difference in the tabulated range is about 1.7% near $z \sim 0.5$.

z	α_B	D_{kin}	Ω_T
10	2.07×10^{-7}	1.64×10^{-6}	0.00194
3	8.60×10^{-5}	6.80×10^{-4}	0.03915
2.1	3.72×10^{-4}	0.00294	0.08060
1.5	0.00122	0.00959	0.14321
1.0	0.00380	0.02988	0.24551
0.8	0.00622	0.04889	0.30787
0.5	0.01346	0.10536	0.43200
0.3	0.02271	0.17740	0.53538
0	0.04940	0.38485	0.70791

Table 1: Reference-branch background evolution.

13.1 Prospective growth target

In the minimal high- n realization, the direct topographic fifth-force term is negligible on ordinary RSD modes and the dominant growth shift is produced by the altered background history. With identical matter-era normalization,

$$R_{f\sigma_8}(z) = \frac{f\sigma_{8,\text{R1}}}{f\sigma_{8,\Lambda\text{CDM}}}$$

is

z	$R_{f\sigma_8}$	suppression
2.1	0.99346	0.65%
1.5	0.98869	1.13%
1.0	0.98164	1.84%
0.8	0.97795	2.20%
0.5	0.97271	2.73%
0.3	0.97168	2.83%
0.0	0.98482	1.52%

Table 2: R1 growth forecast relative to matched curved ΛCDM . These values are reference-branch predictions and become formally preregistered only when reproduced by the canonical code and written to the hashed manifest.

The central phenomenological result is therefore not a large generic modification of gravity. It is the combination

enhanced horizon-scale $n = 1$ susceptibility
+ rapid high- n recovery
+ few-percent late-time growth shift

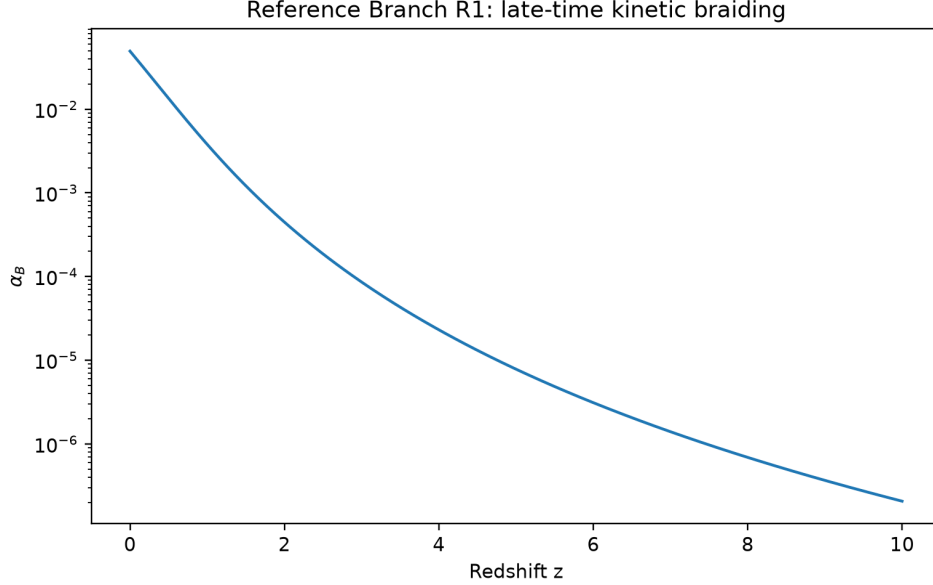


Figure 1: Kinetic-braiding function $\alpha_B(z)$ for Reference Branch R1. The logarithmic vertical scale emphasizes the suppression of the braiding sector toward high redshift.

14 Reference-Branch Figures

15 Discussion

The closed-FLRW gauge audit produces a concrete correction. The first nonconstant scalar harmonic in our original notation, $n = 1$, maps to the exceptional $N = 2$ sector of standard closed-universe perturbation theory. Its scalar-derived trace-free tensor harmonic vanishes, and the regular generally covariant one-field scalar sector contains no propagating scalar degree of freedom there. Modern closed-inflation calculations likewise take physical scalar modes to begin at $N \geq 3$ [2, 3].

The natural response is to move the topographic carrier to the first physical scalar level: our $n = 2$, equivalent to $N = 3$. A degree-two harmonic on S^3 is a traceless quadratic form in the $\mathbb{R}^4$ embedding coordinates. Its cross components A_{4i} generate an exact observer-sky dipole

$$T_{2,\text{dip}} \propto \sin 2\psi (\hat{\mathbf{n}} \cdot \hat{\mathbf{p}})$$

without requiring a sky quadrupole.

The reduced-response filter also adapts naturally. The corrected window

$$\frac{1}{15} < \ell^2 < \frac{1}{8}$$

softens $n = 2$ while stiffening every higher propagating scalar harmonic $n \geq 3$. The formal $n = 1$ coefficient is irrelevant to the regular physical scalar spectrum. The enhanced carrier is therefore the first physical compact-space scalar mode.

The role of BHSM remains structural rather than a completed microscopic derivation. The imported element is the constraint-reduced response logic. The specific mixed block and the source law S_2 remain exploratory until derived from the parent boundary action.

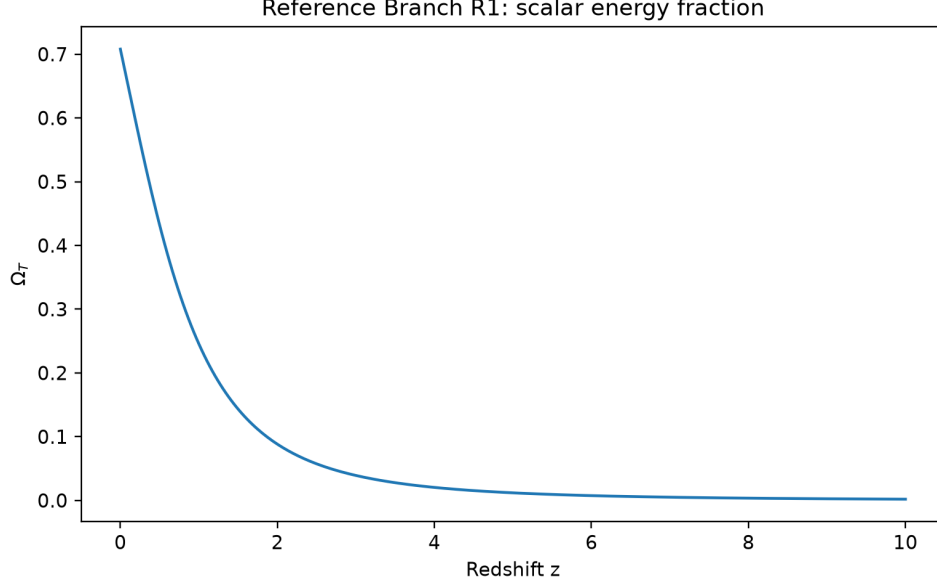


Figure 2: Scalar energy fraction $\Omega_T(z)$ for R1. The scalar component is subdominant at high redshift and becomes dynamically important only at late times.

The complete $n = 2$ scalar–metric transfer kernel must still be derived in the closed cubic-Galileon background. $D_{\text{kin}} > 0$ remains an inherited kinetic diagnostic rather than a complete closed-background stability proof. R1 values $q = 3$ and $\lambda = 1$ remain reference choices, and the frozen curvature forecast must not be retuned to observational results.

The R1 numerical background and high- n growth forecast do not depend on this harmonic correction. Their frozen manifest is therefore preserved unchanged, while the theoretical amendment is recorded separately before comparison with the targeted independent datasets.

16 Conclusions

The closed-FLRW gauge analysis resolves the principal ambiguity in the original bridge construction. Our $n = 1$ scalar harmonic maps to the exceptional $N = 2$ sector and is not retained as the ordinary propagating scalar carrier. The first physical scalar level is

$$\boxed{n_\star = 2} \quad (\boxed{N_\star = 3}).$$

This first physical harmonic contains an exact sky-dipole subspace,

$$T_{2,\text{dip}} = A \sin[2\psi(z)](\hat{\mathbf{n}} \cdot \hat{\mathbf{p}}),$$

when only the cross components A_{4i} are populated.

The corrected selective-response window is

$$\boxed{\frac{1}{15} < \ell^2 < \frac{1}{8}},$$

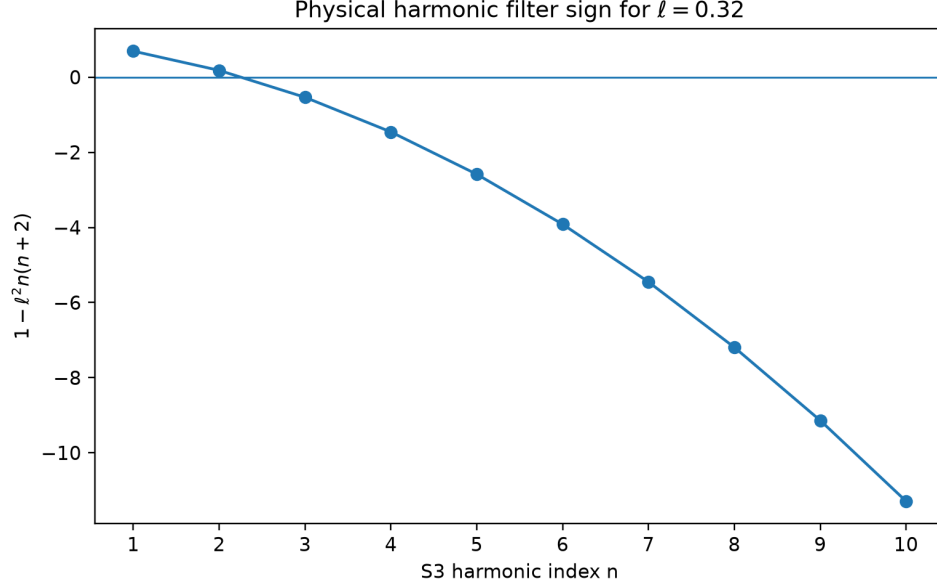


Figure 3: Sign of the Schur-response factor $1 - \ell^2 n(n+2)$ for the illustrative filtering value $\ell = 0.40$. The $n = 1$ branch lies on the softening side while every displayed $n \geq 2$ mode lies on the stiffening side.

within which $n = 2$ is softened and every higher propagating mode $n \geq 3$ is stiffened. The next-level source-normalized response obeys

$$\left| \frac{T_3/S_3}{T_2/S_2} \right| = \frac{8C_2}{15C_3} < \frac{8}{15}.$$

The carried-forward supernova amplitude now calibrates the physical $n = 2$ dipole response. The R1 background and high- n growth predictions remain unchanged.

The revised signature is therefore

first physical S^3 scalar mode ($n = 2$) enhanced
 $\oplus$ fixed-axis dipole subspace with $\sin 2\psi$ envelope
 $\oplus$ suppressed $n \geq 3$ susceptibility
 $\oplus$ near-GR high- k growth.

The next calculation is the full gauge-invariant $n = 2$ scalar-metric transfer matrix in the closed cubic-Galileon background.

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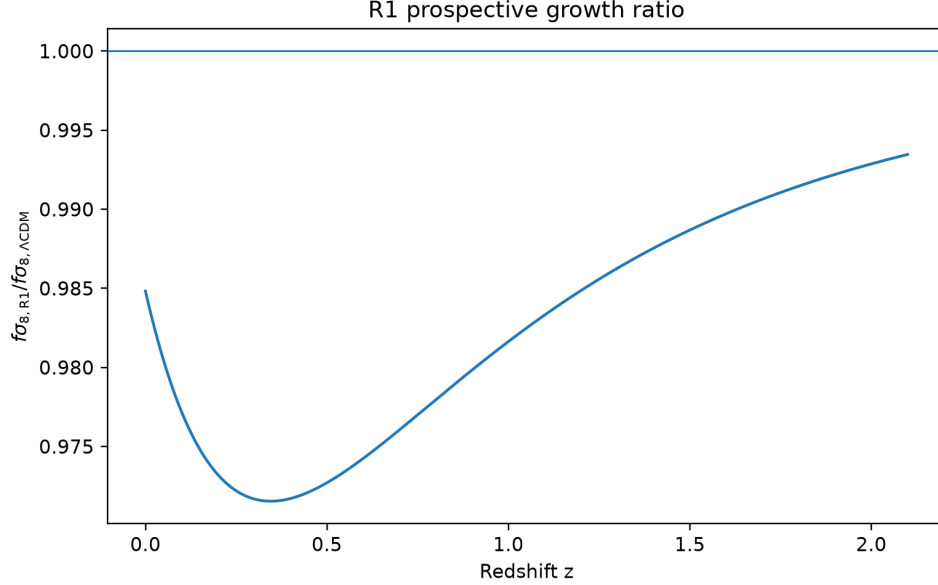


Figure 4: Prospective R1 growth ratio relative to a matched curved- Λ CDM background with the same early normalization.

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